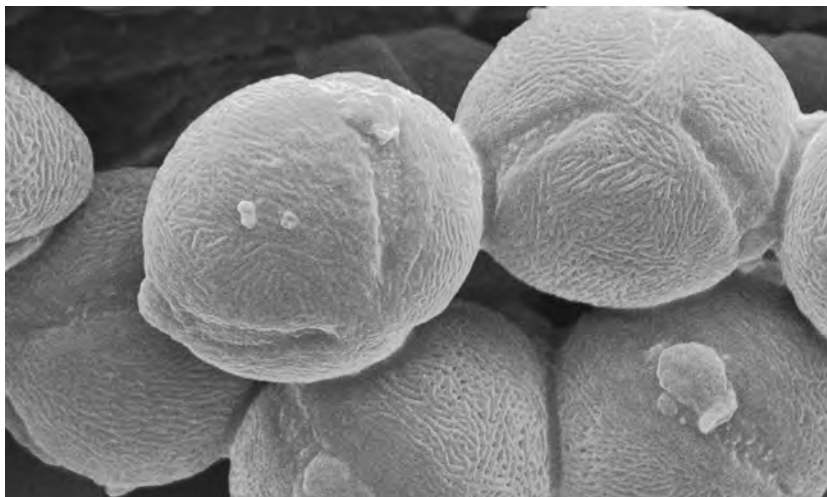


- 2 The photograph shows some pollen grains, as seen using an electron microscope.



(Source: BSIP SA / Alamy Stock Photo)

Scale bar |—————|
20 μm

- (a) Calculate the magnification of this photograph.

Give your answer to **two** significant figures.

(2)

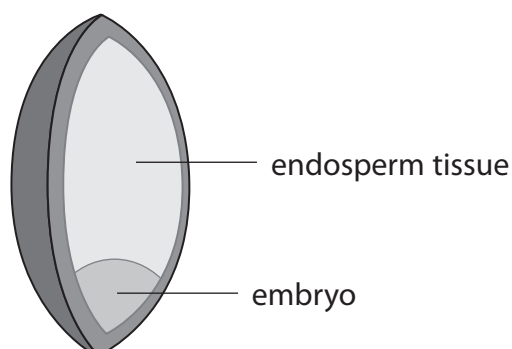
Answer



(b) A pollen grain can produce a pollen tube that contains a generative nucleus.

Plant fertilisation results in the production of an embryo and endosperm tissue.

The diagram shows the location of an embryo and endosperm tissue in a seed.



Describe how the generative nucleus results in the production of an embryo and endosperm tissue in a seed.

(3)

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(c) Flower colour and pollen grain size are genetically determined in foxglove plants.

The linkage of the genes for flower colour and pollen grain size has been investigated.

(i) State what is meant by the term **linkage** with reference to these two genes.

(1)

(ii) A statistical test can be used to analyse observed phenotype frequencies and expected phenotype frequencies for the foxglove plants.

Which of the following statistical tests would be used?

(1)

- ☐ **A** Chi squared
- ☐ **B** correlation coefficient
- ☐ **C** standard deviation
- ☐ **D** Student's t-test

(Total for Question 2 = 7 marks)

